

Hao-Yuan He (何浩源)



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EDUCATION

School of Computer Science, Nanjing Tech University B.Eng. in Computer Science.	2017 – 2021
School of Artificial Intelligence, Nanjing University M.Eng. in Computer Science, LAMDA Group. Advised by Prof. Ming Li.	2021 – 2024
School of Artificial Intelligence, Nanjing University Ph.D. in Computer Science, LAMDA Group. Advised by Prof. Ming Li.	2024 – Now

WORK EXPERIENCE

Research Intern, AI Lab, Tencent Worked on game AI for Honor of Kings (王者荣耀), migrating the training pipeline from TensorFlow 1.12 to PyTorch and developing novel algorithms for fine-grained, competition-level agents.	2023.05 – 2023.07 Shenzhen, China
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RESEARCH

I study reliable neuro-symbolic AI under weak supervision and structured reasoning. Selected publications:

Learning-Augmented Smooth Integer Programs with PAC-Learnable Oracles Developed a learning-augmented framework for NP-hard optimization problems such as MAX-CUT and MAX-k-SAT. The method uses predictions to build a tractable surrogate objective, remains robust to prediction errors, extends approximation guarantees from dense to near-dense settings, and proves the oracle can be learned efficiently from data.	2026
A Learnability Analysis on Neuro-Symbolic Learning Established a characterization of when neuro-symbolic tasks are truly learnable by linking them to their underlying constraint structure.	NeurIPS'2025 (Spotlight)
Ambiguity-Aware Abductive Learning Proposed an ambiguity-aware approach for abductive learning when logical reasoning returns multiple plausible supervision candidates. Instead of committing to one candidate too early, the method models uncertainty over candidates, yielding more reliable supervision and stronger performance.	ICML'2024

ACADEMIC SERVICE

Conference Reviewer / PC Member Reviewed for ACML, IJCLR, ICLR, ICML, NeurIPS, and AAAI.	2023 – Now
Journal Reviewer Reviewed for Knowledge and Information Systems.	2023 – Now

AWARDS

Outstanding Graduate Student Nanjing University	2023, 2024
Winner International Algorithm Case Competition, Track of Data Selection and Label Correction	2022
First Prize of Academic Scholarship Top 10%, Nanjing University	2022, 2023
National Encouragement Scholarship Jiangsu Education Department	2019
First Prize, Chinese Mathematics Competitions Chinese Mathematical Society	2018, 2020

PUBLICATIONS

[1] [H.-Y. He](#) and M. Li, “A Learnability Analysis on Neuro-Symbolic Learning,” in *NeurIPS*, 2025.

- [2] [H.-Y. He](#), Y. Liu, R.-B. Liu, Z. Xie, and M. Li, “Probabilistic Instance Dependent Label Refinement for Noisy Label Learning,” *Machine Learning*, 2025.
- [3] [H.-Y. He](#), H. Sun, Z. Xie, and M. Li, “Ambiguity-Aware Abductive Learning,” in *ICML*, 2024.
- [4] Y.-X. Huang, [H.-Y. He](#), and M. Li, “Tri-ABL: Enhancing ABL Using Three Learners,” in *PAKDD*, 2026.
- [5] [H.-Y. He](#), W.-Z. Dai, and M. Li, “Reduced Implication-bias Logic Loss for Neuro-Symbolic Learning,” *Machine Learning*, 2024.
- [6] Z. Xie, Y. Liu, [H.-Y. He](#), M. Li, and Z.-H. Zhou, “Weakly Supervised AUC Optimization: A Unified Partial AUC Approach,” *TPAMI*, 2024.
- [7] H. Sun, Z. Xie, [H.-Y. He](#), and M. Li, “Mitigating Negative Transfer via Reducing Environmental Disagreement,” *TPAMI*, 2026.
- [8] [H.-Y. He](#) and M. Li, “Learning-Augmented Smooth Integer Programs with PAC-Learnable Oracles,” *Arxiv*.